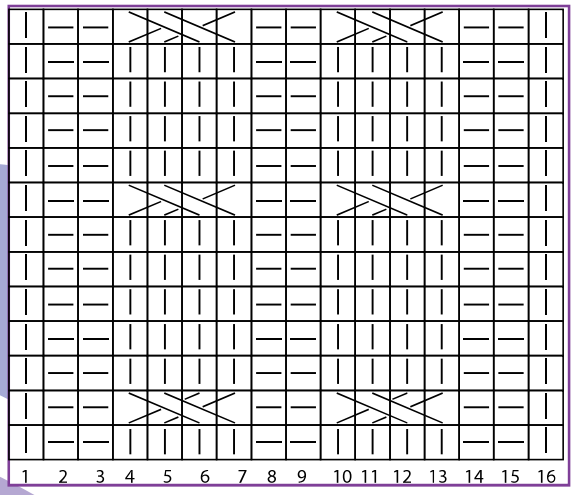
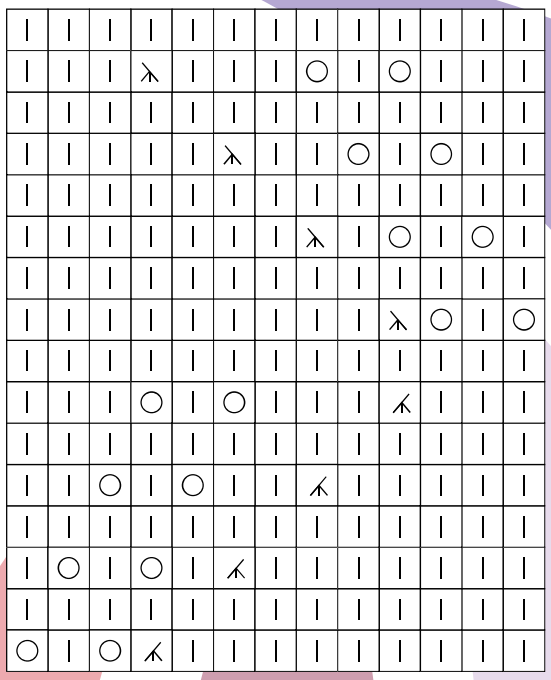
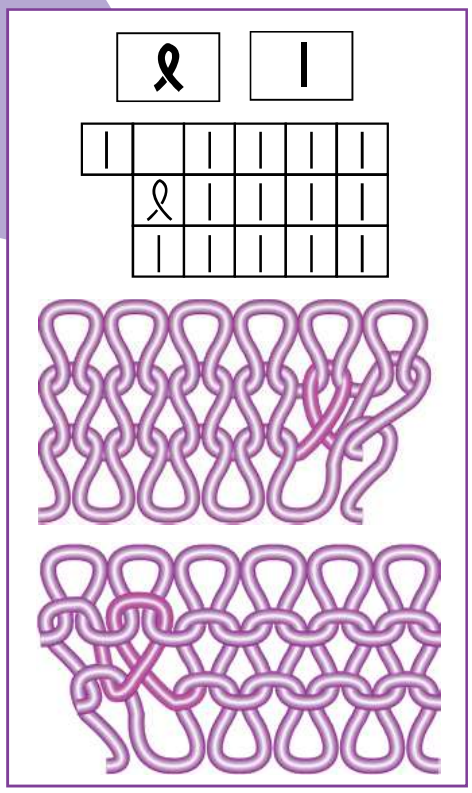
Hand Knit to Machine Knit

Stitch Symbols Unraveled

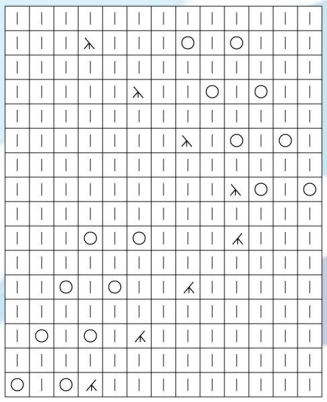






Hand Knit to Machine Knit

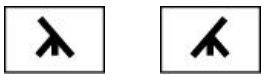
Stitch Symbols Unraveled



Reading knitting charts for both hand and machine knitting doesn't have to be mysterious.

- Start simple and learn what each of the common symbols mean.
- Knit your gauge swatch entirely from the chart for practice
- Generally charts are read from bottom and show the work from the right (public) side
- Use sticky notes or highlighter tape to keep your place
- Try reading charts for hand manipulated stitches before you try programming your machine.
- The symbols used here are shown as viewed from the public side of the garment.

For automatic patterning machines (punchcard and electronic), it is critical that you understand the workings of YOUR machine.



Machine knitters need to keep in mind that we are looking at our work from the purl side. Charts created for hand knitting need to be reversed.

Examine the stitches used and imagine how they will look on the public side of the work.

If your hand knitting chart has a left leaning increase on the left side of the piece, you will be transferring stitches right on the right side of the piece.

Don't let the words "left" and "right" get you confused. Create a mental picture of the finished item before you start to knit.

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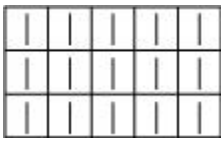
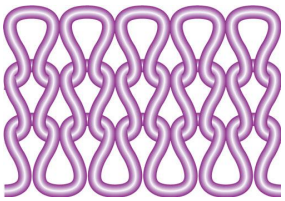
Hand Knit to Machine Knit

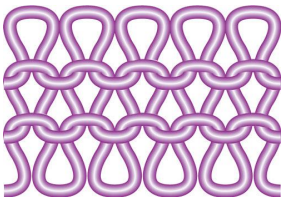
Stitch Symbols Unraveled

Knit Stitch





Loop of plain stitch (public side)



Private Side

The smooth side is considered the "right" (public) side.

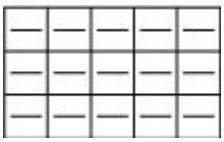
Machine Knitting

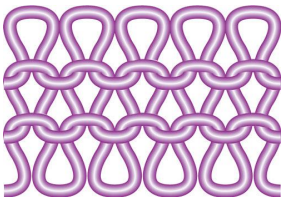
Also called stocking stitch, stockinette stitch or plain knitting this is the default stitch for machine knitters.

Hand Knitting (k)

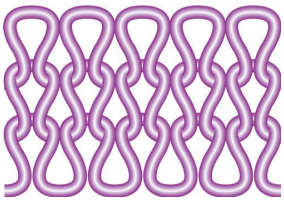
Produced by knitting one row (right side) and purling one row (wrong side).

Purl Stitch





Reverse Stockinette public side



Private side

The bumpy side is considered the "right" (public) side.

Machine Knitting

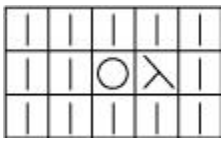
Referred to as Reverse Stockinette Stitch.

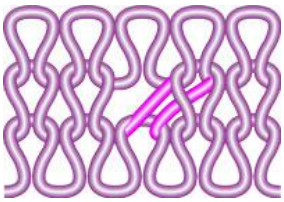
Combine this with plain knitting by removing the work from the machine and turning it, re-hanging each stitch. The next row knitted will be reversed.

Hand Knitting (p)

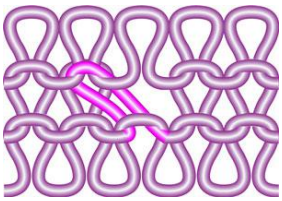
Produced by purling one row (right side) and knitting one row (wrong side).

Eyelet





Eyelet = Hole (public side)



Eyelet = Hole (private side)

Intentional holes in your knitting – lace

Machine Knitting

A stitch is transferred to another needle and the emptied needle remains in working position. On the next pass of the carriage a new stitch is formed.

Hand Knitting (yo)

Stitches are knit together (decreases) and a yarn over is worked in the row, creating an eyelet and maintaining the correct stitch count for the row.

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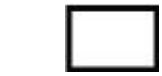
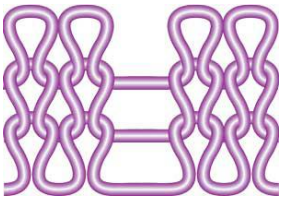
Hand Knit to Machine Knit

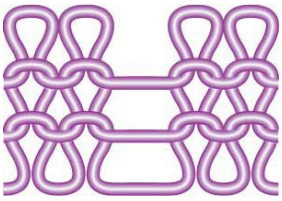
Stitch Symbols Unraveled

Needle out of Work





Column of "un-knit" stitches



Private Side

Machine Knitting:

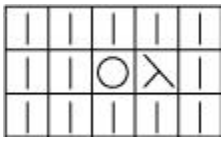
When a needle is left in out of work position, the yarn is carried between stitches, forming a loose, opening.
Columns may continue throughout the entire knitted piece.

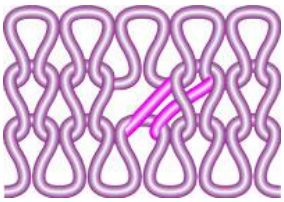
After creating an eyelet, leave the emptied needle out of work for a few rows for a decorative effect.

Hand Knitting:

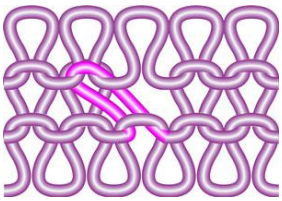
No exact equivalent is available for hand knitter's. A yarn over (YO) may be worked on one row, then dropped on the next, creating an open column.

Two Stitches Together (Right)





Right Slant on Public Side



Machine Knitter's View

Machine Knitting:

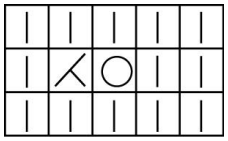
Transfer a single stitch to the needle to the LEFT.
When viewed from the public side, the stitch slants RIGHT.
Keep the emptied needle in working position.

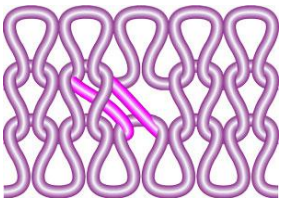
Hand Knitting:

SSK, YO

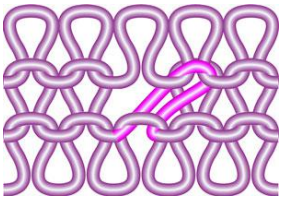
Slip, slip, knit; then yarn over

Two Stitches Together (Left)





Left Slant on Public Side



Machine Knitter's View

Machine Knitting:

Transfer a single stitch to the needle to the RIGHT.
When viewed from the public side, the stitch slants LEFT.
Keep the emptied needle in working position.

Hand Knitting:

YO, k2tog

Yarn over, knit 2 together

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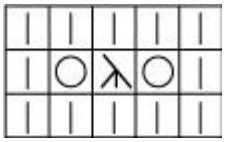
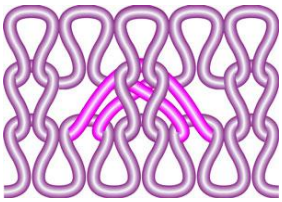
Hand Knit to Machine Knit

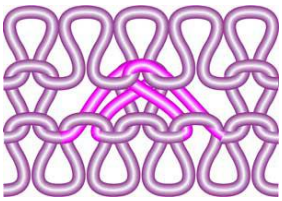
Stitch Symbols Unraveled

Three Stitches Together (Center)





Center Stitch on top (public side)



Machine Knitter's View

Machine Knitting

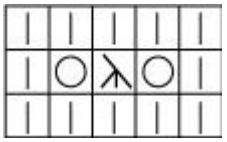
Transfer 1 stitch right and one stitch left. 3 stitches on the center needle. Keep emptied needles in working position.

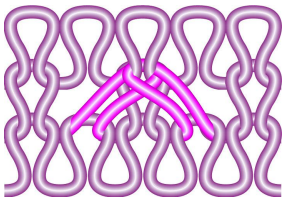
Hand Knitting (s2kp2)

Sl2, k1, p2sso

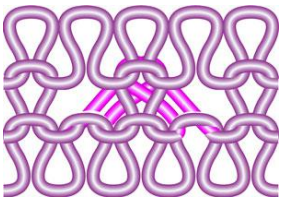
Slip two stitches knitwise, knit 2 toget1her, pass the slipped stitches over

Three Stitches Together (Right on top)





Right Stitch on top (public side)



Machine Knitter's View

Machine Knitting

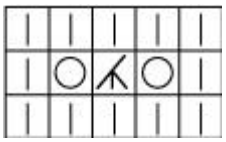
With a single transfer tool, remove the center stitch and hold it.
Using a separate single transfer tool, transfer the left stitch to the emptied center needle, then transfer the right stitch to the center needle.
Return the held stitch to the center needle (3 stitches on the center needle). Keep the emptied needles in working position.

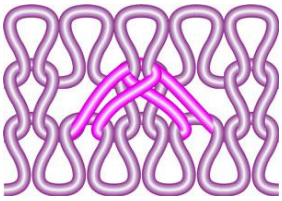
Hand Knitting (sk2p)

Sl1,k2tog, psso

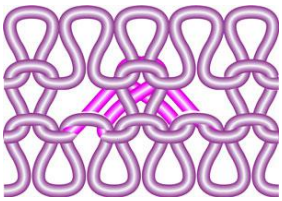
Slip one, knit 2 together, pass the slipped stitch over

Three Stitches Together (Left on top)





Left Stitch on top (public side)



Machine Knitter's View

Machine Knitting

With a single transfer tool, remove the center stitch and hold it.
Using a separate single transfer tool, transfer the right stitch to the emptied center needle, then transfer the left stitch to the center needle.
Return the held stitch to the center needle (3 stitches on the center needle). Keep the emptied needles in working position.

Hand Knitting (k3tog)

k3tog

Knit 3 together
(Not exactly like the MK version, but the front stitch slants correctly)

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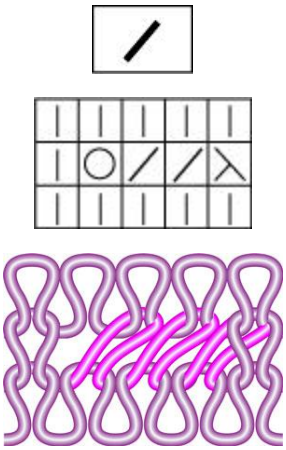
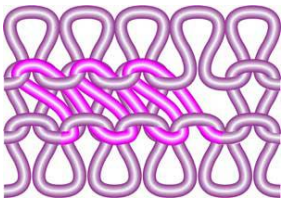
Hand Knit to Machine Knit

Stitch Symbols Unraveled

Transfer Right



3 stitches leaning right



Machine knitter's view

Often used for knitting lace

Machine Knitting

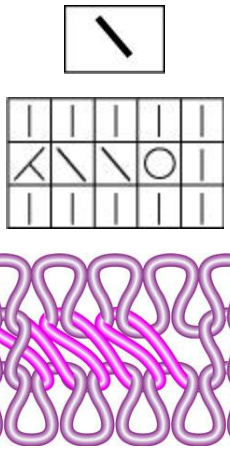
Using a multi-prong transfer tool, move 2-3 (or more) stitches one needle to the LEFT. There will be 2 stitches on the last needle and one emptied needle.
Keep the emptied needle in working position to form an eyelet.
On the public side the stitches will lean RIGHT.

Hand Knitting

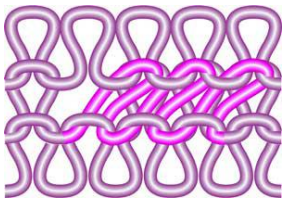
k2tog, k2, YO

Knit 2 together (right slanting), knit 2, yarn over

Transfer Left



3 stitches leaning left



Machine knitter's view

Often used for knitting lace

Machine Knitting

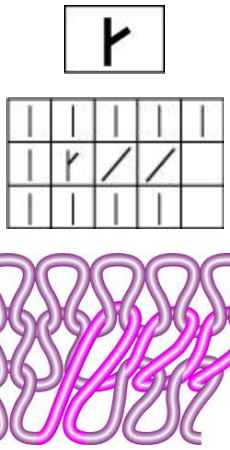
Using a multi-prong transfer tool, move 2-3 (or more) stitches one needle to the RIGHT. There will be 2 stitches on the last needle and one emptied needle.
Keep the emptied needle in working position to form an eyelet.
On the public side the stitches will lean LEFT.

Hand Knitting

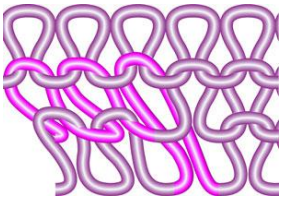
YO,k2,ssk

Yarn over, knit 2, slip, slip, knit both stitches. (left slanting)

Right Leaning Increase



2 stitch increase right



Machine knitter's' view

Traditional machine knitting increase

Machine Knitting

Using a multi-prong transfer tool, move 2-3 (or more) stitches one needle to the LEFT. There will be 2 stitches on the last needle and one emptied needle.
Pick up the purl bump from the row below and place it on the emptied needle.
On the public side the stitches will lean RIGHT.

Hand Knitting

This technique is not used in hand knitting – see "Make 1 (m1)"

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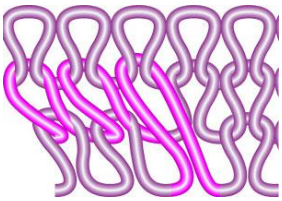
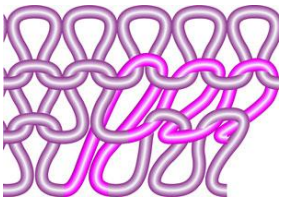
Hand Knit to Machine Knit

Stitch Symbols Unraveled

Left Leaning Increase



2 stitch increase left



Machine knitter's view

Traditional machine knitting increase

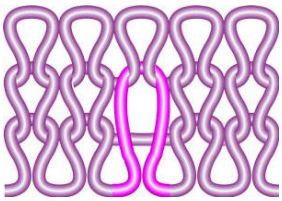
Machine Knitting

Using a multi-prong transfer tool, move 2-3 (or more) stitches one needle to the RIGHT. There will be 2 stitches on the last needle and one emptied needle.
Pick up the purl bump from the row below and place it on the emptied needle.
On the public side the stitches will lean LEFT.

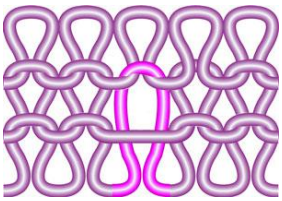
Hand Knitting

This technique is not used in hand knitting – see "Make 1 (m1)"

Slip (Skip) Stitch



Elongated stitch



Strand on purl side

A stitch is pulled up without knitting and yarn is carried on the PURL side. The stitch can be carried over multiple rows and multiple stitches may be slipped for texture on the reverse side of the work. Color changes can also be incorporated.

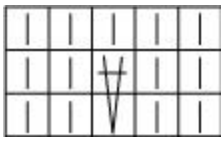
Machine Knitting

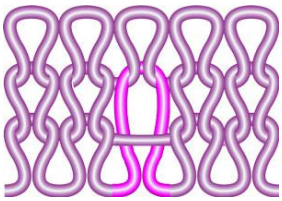
See your machine manual for slip stitch settings

Hand Knitting

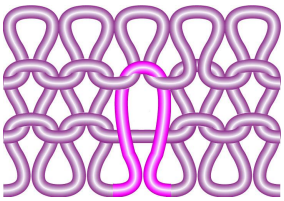
Often called "Mosaic Knitting"

Float Stitch





Strand on Knit side



Strand on purl side

A stitch is pulled up without knitting and yarn is carried on the KNIT side.

The stitch can be carried over multiple rows and multiple stitches may be slipped for texture on the reverse side of the work. Color changes can also be incorporated.

Machine Knitting

Manually reverse the stitch, so the float is carried on the knit side of the work.

Hand Knitting

Not often seen in hand knitting

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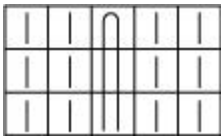
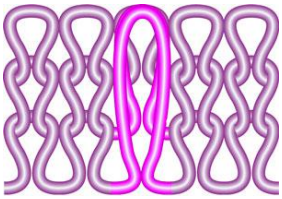
Hand Knit to Machine Knit

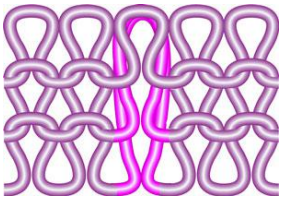
Stitch Symbols Unraveled

Tuck Stitch





Loop(s) over needle



Machine knitter's view

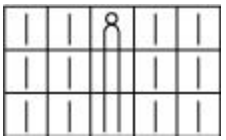
Machine Knitting

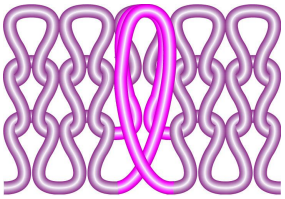
Traditional machine knitting stitch. Can be worked on automatic or non-patterning machines. Stitch is not knit and loop is carried OVER the patterning needle, creating texture.
Can be worked over multiple rows.
Automatic tucking over adjacent stitches is problematic.

Hand Knitting

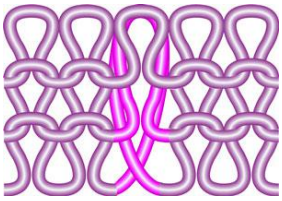
A variation is "knitting in the row below"

Twisted Tuck Stitch





Manually twisted



Machine knitter's view

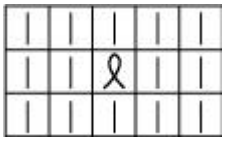
Machine Knitting

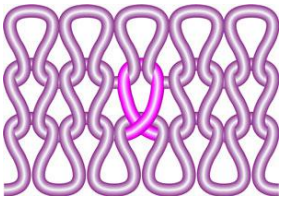
Tucked strands are manually twisted before knitting. This gives additional texture

Hand Knitting

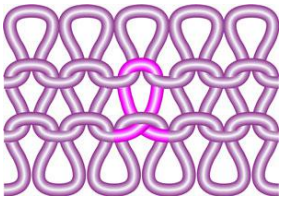
This technique is not used in hand knitting

Twisted Stitch





Manually twisted



Machine knitter's view

Machine Knitting

Individual stitches or rows of stitches are manually twisted. Use a latch tool and remove a single stitch, twist 90 degrees and re-hang the stitch on the needle.
Gives subtle texture

Hand Knitting

Knit or purl in the back loop. This twists the stitch for an interesting texture.

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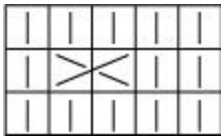
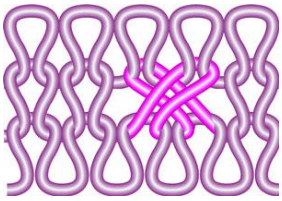
Hand Knit to Machine Knit

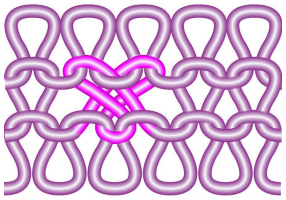
Stitch Symbols Unraveled

Crossing Left over Right





Mini cable crossing



Machine knitter's view

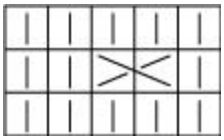
Machine Knitting

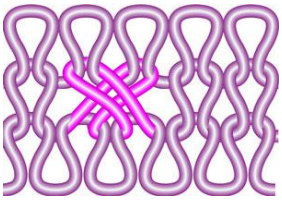
Using 2 single prong transfer tools, remove 2 stitches from their needles. Cross RIGHT stitch over Left and return each stitch to the opposite needle. On the public side of the knitting, the left stitch will cross OVER the right.

Hand Knitting

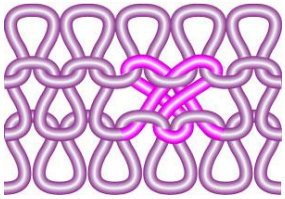
Knit 1 stitch. Slip the next stitch to a stitch holder and hold in BACK. Knit the next stitch. Knit the stitch from the stitch holder.

Crossing Right over Left





Mini Cable Crossing



Machine knitter's view

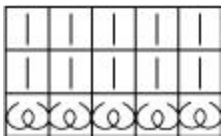
Machine Knitting

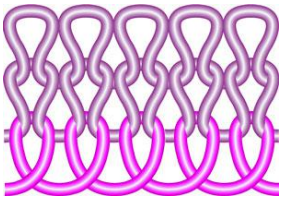
Using 2 single prong transfer tools, remove 2 stitches from their needles. Cross LEFT stitch over Right and return each stitch to the opposite needle. On the public side of the knitting, the right stitch will cross OVER the left.

Hand Knitting

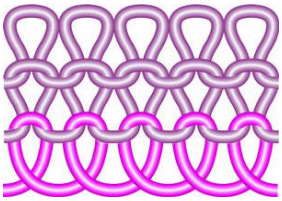
Knit 1 stitch. Slip the next stitch to a stitch holder and hold in FRONT. Knit the next stitch. Knit the stitch from the stitch holder.

Winding Stitch





Public (Knit) side



Private (Purl) Side

Machine Knitting

Commonly used cast on method also known as e-wrap. Wrap the yarn around each needle (as the letter e). Shown as worked from left to right, the cast on can be worked from right to left.

This is a loose cast on that is often used when adding a few stitches to the side of your knitting.

Hand Knitting

Known as the thumb, single loop or backward loop cast on. A strand of yarn is wound around the knitting needle providing a loose edge.

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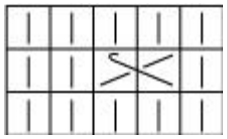
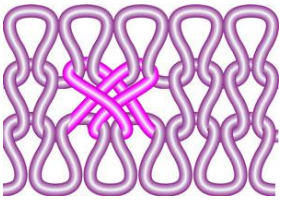
Hand Knit to Machine Knit

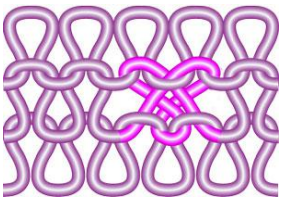
Stitch Symbols Unraveled

Crossed Stitch Left





Left stitch through right



Machine knitter's view

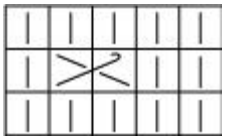
Machine Knitting

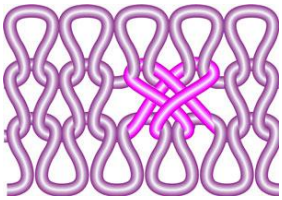
Using 2 single prong transfer tools, remove 2 stitches from their needles. Insert a latch tool through the LEFT stitch, keeping the stitch on both tools. With the hook grab the RIGHT stitch and pull it through, removing the latch tool.
Rehang each stitch, crossing them.

Hand Knitting

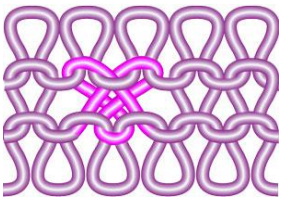
This is not a technique generally used in hand knitting.

Crossed Stitch Right





Right stitch through left



Machine knitter's view

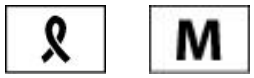
Machine Knitting

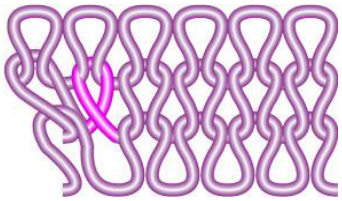
Using 2 single prong transfer tools, remove 2 stitches from their needles. Insert a latch tool through the RIGHT stitch, keeping the stitch on both tools. With the hook grab the LEFT stitch and pull it through, removing the latch tool.
Rehang each stitch, crossing them.

Hand Knitting

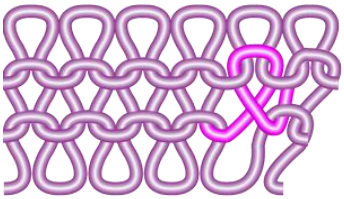
This is not a technique generally used in hand knitting.

Make 1 Left





Twisted increase left leaning



Machine knitter's view

Machine Knitting

Transfer 1 or more stitches 1 needle to the RIGHT. Use a latch tool and pick up the strand below the emptied needle. Twist the strand and re-hang it on the emptied needle. Twist so the strand on the public side of the work leans LEFT.

Hand Knitting (m1 left)

Insert the left-hand needle from front to back under the strand of yarn between the stitches on the row below. Knit this stitch through the back loop.

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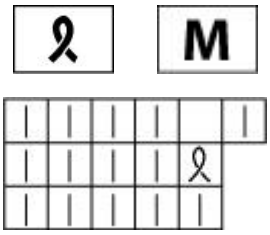
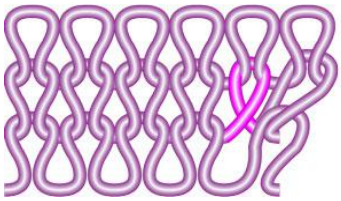
Hand Knit to Machine Knit

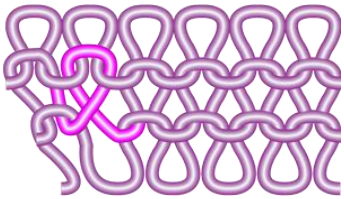
Stitch Symbols Unraveled

Make 1 Right





Twisted increase right leaning



Machine knitter's view

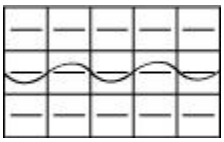
Machine Knitting

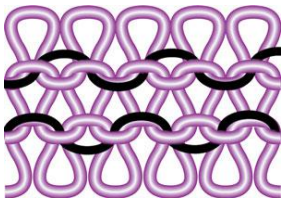
Twisted increase.
Transfer 1 or more stitches 1 needle to the LEFT. Use a latch tool and pick up the strand below the emptied needle. Twist the strand and re-hang it on the emptied needle. Twist so the strand on the public side of the work leans RIGHT.

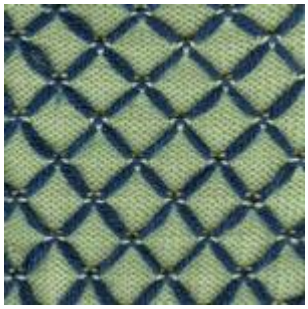
Hand Knitting (m1 right)

Insert the left-hand needle from front to back under the strand of yarn between the stitches on the row below. Knit this stitch through the front loop.

Weaving







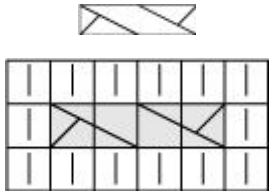
Machine Knitting

Knit Weave
Weave strands of yarn over and under the needles before passing the carriage to knit a row. Use novelty and heavier yarn for amazing results. Weave over multiple needles, creating patterns. Patterns show in the purl side of the knitting. Automatic patterning machines speed up the needle selection, but this technique can be worked on any machine.

Hand Knitting

This technique is not used by hand knitters.

Cable Left



Machine Knitting

Machine knitters can create classic cables by manually removing, transferring and crossing individual or groups of stitches.

Reading stitch charts is the easiest way to follow a cable pattern.

See the following tips for cables on the knitting machine.

Hand Knitting (2/2 LC)

Creating cables by counting stitches and reading written text or by using a chart is personal preference.

2/2 LC (2 stitch left cross)

Sl 2 to cn, hold to front, k2;k2 from cn

Slip 2 to cable needle, hold in front, knit 2; knit 2 from cable needle

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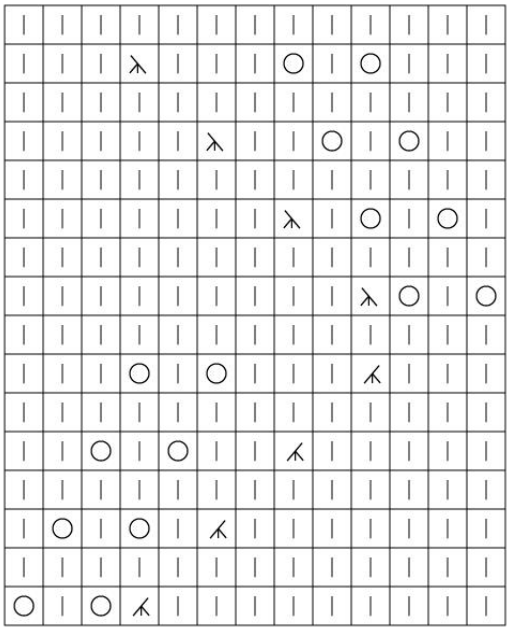
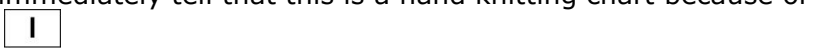
LACE



Row 1: K9, k3tog, yo, k1, yo
Row 2 and ALL WS Rows: Purl
Row 3: K7, k3tog, k1, you, k1, yo,k1.
Row 5: K5, k2tog, k2, yo, k1, yo, k2.
Row 7: k3, k3tog, k3, yo,k1,yo,k3.
Row 9: Yo, k1, yo, s2kp, k9.
Row11:K1, yo, k1, yo, k2, s2kp, k5.
Row 15: K3, yo, k1, yo, k3, s2kp, k3

To use a text only pattern for the machine, it is much easier to chart out the pattern to follow as you are knitting.

- You can immediately tell that this is a hand knitting chart because of the knit background stitches. 
- For machine knitting, you'll need to reverse the direction of the transfers.
- Notice that the yarn over (eyelet) symbols are not always directly on either side of the 3 stitches together . For machine knitter's, this indicates that stitches need to be transferred so there are emptied needles (remaining in working position) where the eyelet symbols are shown.

This sequence would indicate (from left to right):
emptied needle, one stitch, emptied needle, 3 stitches on one needle.

Automatic patterning for lace for punchcard and electronic machines is a matter of programming the machine to operate the lace carriage.

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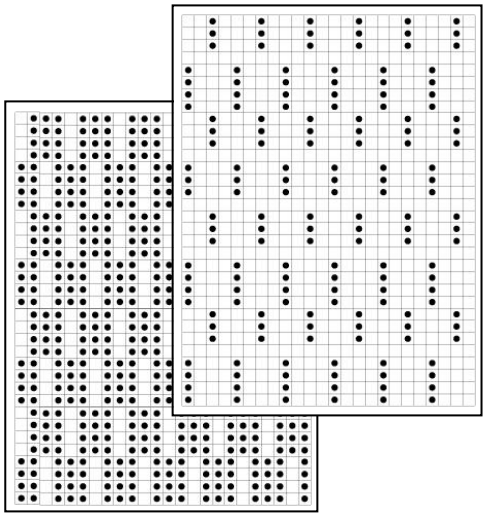
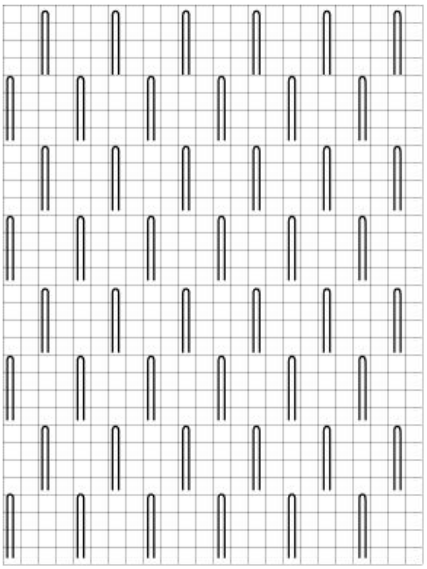
TUCK STITCH



- You won't often see tuck stitches used for hand knitting
- Older machine knitting patterns might use one of these styles of charts for tuck.
- Use the manual for your punchcard or electronic machine to determine the correct format for tuck. Before programming or punching a card, examine a stitch pattern that came with your machine to determine the format.
- If you need to punch or program a large number of black squares, consider using the lower format and the "reverse" setting on your machine.
- We highly recommend Tami Nobuyuki's "Machine Knitting Workbook #3 – Pattern Knitting techniques" for a complete discussion of creating punchcards for tuck (and other stitch types) for various models of machines.



- This symbol indicates the number of rows to tuck. The example is a 4 row tuck. 4 strands will be over the selected needle before all 4 are knitted together.

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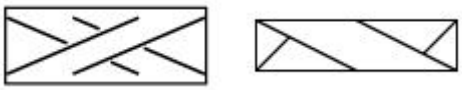
CABLES



For machine knitters, cables are best worked by following a chart. The biggest consideration is to determine if the chart is shown from the public side or the private side. Notice the background stitches. If the majority of the background stitches are knit stitches, the pattern was written for hand knitting. If you want to duplicate the direction of the cables, you will need to reverse them for the machine.

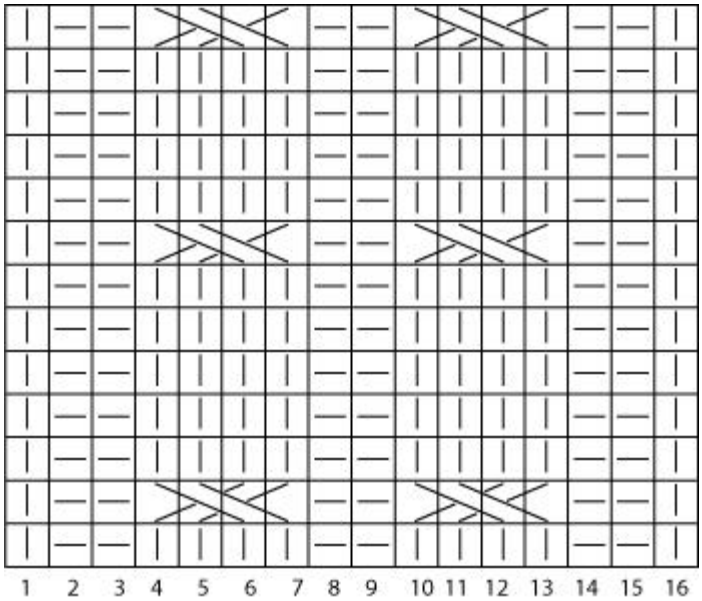
In this example the cables lean to the left. This translates to crossing to the right on our machines. It can be confusing so work a swatch before starting your garment, to make sure the cables are crossing the correct direction.

In this example, for hand knitting, the instructions for the cable row may read:

K1, p2, sl 2 to cn, hold to front, k2;k2 from cn

Since machine knitters aren't counting stitches, (or using cable needles) we refer to needle numbers.

If the pattern you want to knit is written entirely in text, create a simple chart with both stitch and row numbers. Knitting cables is nearly impossible without a chart.



Using a Ribber to Knit Cables

Often a column of purl stitches will be placed on either side of a cable. This gives the cable more depth and definition. For machine knitters this translates to hand manipulating each purl stitch.

Consider putting your ribber to use in creating cabled garments. You can more easily follow hand knitting charts if you place the "purl" stitches on the main bed of the machine and the stitches to be cabled on the ribber. This way, the knit side of the cables is facing you making it easier to cross in the proper direction.

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